T6 — Advanced Organic Chemistry and Analysis

6.1 Aromatic Chemistry — Benzene

Structure of Benzene

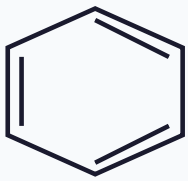
Benzene (C_6H_6) is a planar, cyclic molecule. The Kekule model of alternating single and double bonds is incorrect because:

- All six C-C bonds are the **same length** (0.140 nm), intermediate between single (0.154 nm) and double (0.134 nm) bonds.
- The **enthalpy of hydrogenation** is 152 kJ mol^{-1} less exothermic than expected for three isolated C=C bonds, demonstrating extra thermodynamic stability (**delocalisation energy**).
- Benzene undergoes **electrophilic substitution** rather than addition, preserving the aromatic ring stability.

Delocalised model: Each carbon is sp^2 hybridised with one unhybridised p orbital perpendicular to the ring. The six p orbitals overlap sideways to form a continuous ring of delocalised π electron density above and below the plane, stabilising the molecule.

Benzene: Kekule vs Delocalised Model

Kekule (incorrect)



Predicts 2 bond lengths, addition reactions
ΔH

$_{hyd}$ should be -360 kJ/mol vs Delocalised (correct) All bonds equal, substitution not addition ΔH_{hyd} = -208 kJ/mol (152 kJ

more stable) 6 p-orbitals overlap → delocalised π system → extra thermodynamic stability

Electrophilic Substitution of Benzene

Benzene reacts with electrophiles via substitution (not addition), preserving aromatic stability. The mechanism has three steps: (1) generate the electrophile, (2) π electrons attack electrophile forming a Wheland intermediate (non-aromatic, positively charged), (3) H^+ is lost to restore aromaticity.

Nitration

$$C_6H_6 + HNO_3 \rightarrow C_6H_5NO_2 + H_2O$$

Conditions: Conc. HNO_3 + conc. H_2SO_4, 50°C. **Electrophile:** NO_2^+ (nitronium ion) from $HNO_3 + H_2SO_4 \rightarrow NO_2^+ + HSO_4^- + H_2O$. H_2SO_4 is regenerated (catalyst).

Friedel-Crafts Acylation

$$C_6H_6 + RCOCl \rightarrow C_6H_5COR + HCl$$

Conditions: RCOCl + $AlCl_3$ catalyst, reflux, anhydrous. **Electrophile:** RCO^+ from $RCOCl + AlCl_3 \rightarrow RCO^+ + AlCl_4^-$.

Friedel-Crafts Alkylation

$$C_6H_6 + RCl \rightarrow C_6H_5R + HCl$$

Conditions: RCl + $AlCl_3$. **Electrophile:** R^+. Alkylation can lead to polysubstitution (alkyl groups activate the ring); acylation is preferred for monosubstitution.

Phenol

Phenol (C_6H_5OH) has an -OH directly on the ring. The oxygen lone pair delocalises into the ring, increasing electron density. Consequences:

- **More reactive** towards electrophilic substitution than benzene (reacts with Br_2(aq) at room temperature without a catalyst, giving a white precipitate of 2,4,6-tribromophenol)

- **More acidic** than aliphatic alcohols (pK_a = 10). The phenoxide ion is stabilised by delocalisation. But weaker than carboxylic acids — does NOT react with Na_2CO_3.

Reduction of Nitrobenzene to Phenylamine

$$C_6H_5NO_2 + 6[H] \rightarrow C_6H_5NH_2 + 2H_2O$$

Conditions: Sn + conc. HCl, reflux. Then add excess NaOH to liberate the free amine.

Exam Tip: In electrophilic substitution mechanisms: show the electrophile generation, a curly arrow from the ring π system to the electrophile, the Wheland intermediate with a + charge, and loss of H^+ to restore aromaticity. For Friedel-Crafts, also show regeneration of $AlCl_3$.

6.2 Aldehydes and Ketones

Aldehyde (RCHO): Carbonyl at end of chain (bonded to at least one H).
Ketone (RCOR'): Carbonyl within chain (bonded to two C atoms).
Both contain the polar C=O (δ+C=Oδ-), making the carbon electrophilic and susceptible to nucleophilic addition.

Nucleophilic Addition with HCN

$$RCHO + HCN \rightarrow RCH(OH)CN \quad \text{(hydroxynitrile)}$$

Conditions: NaCN/KCN in dilute H_2SO_4 (generates HCN in situ). CN^- is the nucleophile.

Mechanism: (1) CN^- lone pair attacks δ+ C of C=O, π bond breaks, electrons move to O, forming alkoxide $RCH(CN)O^-$. (2) O^- protonated by HCN → -OH formed, CN^- regenerated.

Key points: Extends carbon chain by 1. Creates a chiral centre → racemic mixture. The -CN can be hydrolysed to -COOH or reduced to $-CH_2NH_2$.

Reduction with $NaBH_4$

$NaBH_4$ provides H^- (hydride) as the nucleophile. Aldehydes → primary alcohols. Ketones → secondary alcohols. Conditions: $NaBH_4$ in water/methanol, room temperature.

Distinguishing Aldehydes from Ketones

Test	Reagent	Aldehyde	Ketone
Tollens'	$[Ag(NH_3)_2]^+$, warm	Silver mirror	No reaction
Fehling's	Cu^{2+}/NaOH, heat	Blue → brick-red Cu_2O precipitate	No reaction
$K_2Cr_2O_7/H^+$	Acidified dichromate, warm	Orange → green	No reaction

2,4-DNPH (Brady's): Both give orange/yellow precipitate (confirms C=O). **Iodoform test:** Positive for CH_3CO- (methyl ketone) and $CH_3CH(OH)$- groups. Yellow precipitate of CHI_3 with NaOH + I_2.

Exam Tip: Identification strategy: (1) 2,4-DNPH confirms C=O. (2) Tollens'/Fehling's distinguishes aldehyde from ketone. (3) Iodoform identifies CH_3CO- specifically. Always state the exact observation.

6.3 Carboxylic Acids and Esters

Carboxylic acids (-COOH) are weak acids: $RCOOH \rightleftharpoons RCOO^- + H^+$. The carboxylate ion is stabilised by delocalisation over both oxygens.

Key reactions: With metals (H_2 evolved), carbonates (CO_2 evolved — effervescence test), bases (neutralisation), alcohols (esterification with H_2SO_4 catalyst).

Esters

Named: alkyl from alcohol + -oate from acid. **Hydrolysis:** Acid hydrolysis (reversible) → acid + alcohol. Base hydrolysis (irreversible, saponification) → sodium salt + alcohol.

6.4 Acyl Chlorides and Acid Anhydrides

Acyl chlorides (RCOCl) react vigorously at room temperature via nucleophilic addition-elimination:

Nucleophile	Products
H_2O	RCOOH + HCl (white fumes)
R'OH	RCOOR' + HCl (ester formed)
NH_3 (excess)	$RCONH_2$ + NH_4Cl (amide)
$R'NH_2$	RCONHR' + $R'NH_3^+Cl^-$ (N-substituted amide)

Advantages over carboxylic acids: faster, irreversible, higher yield, no catalyst needed.

Acid anhydrides undergo the same reactions more gently. By-product is RCOOH (not HCl) — safer, cheaper, preferred industrially.

6.5 Amines

Amine: Organic compound with a nitrogen lone pair. Primary (RNH_2), secondary (R_2NH), tertiary (R_3N). The lone pair makes amines basic (accept H^+) and nucleophilic (donate lone pair to δ+ carbon).

Preparation: (1) RBr + excess NH_3 (sealed tube, heat). (2) RCN + 4[H] ($LiAlH_4$ or H_2/Ni). (3) $ArNO_2$ + 6[H] (Sn/conc. HCl, then NaOH).

Basicity order: Aliphatic amines > NH_3 > phenylamine. Alkyl groups donate electron density to N (+I effect). In phenylamine, the lone pair delocalises into the ring, making it less available.

6.6 Amino Acids and Proteins

General formula: H_2N-CHR-COOH. At the isoelectric point, they exist as zwitterions: ^+H_3N-CHR-COO^-. In acid: cation. In base: anion. All (except glycine) have a chiral centre.

Peptide bonds (-CONH-): Formed by condensation between -COOH of one amino acid and $-NH_2$ of another, with loss of H_2O. Proteins are polypeptides. Hydrolysis: 6M HCl reflux 24h, or enzymes.

6.7 Polymers

Addition Polymers

From alkene monomers (C=C opens). No by-product. 100% atom economy. Non-biodegradable. E.g. poly(ethene), PVC, PTFE.

Condensation Polymers

Polyesters: Diacid + diol → ester linkages (-COO-) + H_2O. E.g. PET.

Polyamides: Diacid + diamine → amide linkages (-CONH-) + H_2O. E.g. Nylon 6,6. Proteins are natural polyamides.

Condensation polymers are more biodegradable than addition polymers (ester/amide linkages can be hydrolysed by enzymes, acid, or base).

> **Exam Tip:** To find monomers from a polymer: locate the linkage (-COO- or -CONH-), break it, add -OH and -H where water was removed. Practise in both directions.

6.8 Organic Synthesis and Retrosynthesis

Retrosynthetic analysis works backwards from the target, identifying the immediate precursor at each step.

Key Inter-conversions

From	Reagent / Conditions	To
Alkene	HBr	Halogenoalkane
Alkene	Steam, H_3PO_4, 300°C	Alcohol
Halogenoalkane	Aq. NaOH, reflux	Alcohol

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Halogenoalkane	Ethanolic KCN, reflux	Nitrile (chain +1)
Halogenoalkane	Ethanolic KOH, hot	Alkene
Halogenoalkane	Excess NH_3, sealed tube	Amine
1° alcohol	$K_2Cr_2O_7/H_2SO_4$, distil	Aldehyde
1° alcohol	$K_2Cr_2O_7/H_2SO_4$, reflux	Carboxylic acid
2° alcohol	$K_2Cr_2O_7/H_2SO_4$	Ketone
Alcohol	Conc. H_2SO_4, heat	Alkene
Aldehyde/ketone	$NaBH_4$	Alcohol
Aldehyde/ketone	HCN/KCN	Hydroxynitrile (chain +1)
Nitrile	Dilute acid, reflux	Carboxylic acid
Nitrile	$LiAlH_4$/dry ether	Primary amine
Acid + alcohol	H_2SO_4 catalyst, reflux	Ester
Acyl chloride + alcohol	Room temperature	Ester + HCl
Benzene	HNO_3/H_2SO_4, 50°C	Nitrobenzene
Nitrobenzene	Sn/HCl, then NaOH	Phenylamine
Benzene	$RCOCl/AlCl_3$	Aryl ketone

Exam Tip: Always state reagents AND conditions. If the target has more carbons, use chain extension (KCN, Friedel-Crafts, or HCN). Work backwards from the target for multi-step synthesis.

6.9 Mass Spectrometry

Molecular ion peak (M^+): Gives relative molecular mass. $M \rightarrow M^{+\bullet} + e^-$.

Fragmentation: M^+ breaks into fragments. m/z values help deduce structure.

m/z loss	Fragment lost
15	$CH_3\bullet$
17	$OH\bullet$
18	H_2O
28	CO or C_2H_4
29	$CHO\bullet$ or $C_2H_5\bullet$
31	$CH_2OH\bullet$
45	$OC_2H_5\bullet$
77	$C_6H_5\bullet$ (phenyl)

TOF MS stages: (1) Ionisation (electron impact or electrospray). (2) Acceleration (all ions get same KE). (3) Drift (lighter ions faster). (4) Detection. Electrospray gives MH^+ (M+1), so subtract 1.

M+1 peak: Due to ^{13}C. Height as % of M ≈ 1.1 × number of C atoms.

6.10 Infrared Spectroscopy

Bond	Wavenumber / cm^{-1}	Appearance	Found in
O-H (alcohol)	3230-3550	Broad	Alcohols, phenols
O-H (carboxylic acid)	2500-3300	Very broad	Carboxylic acids

N-H	3300-3500	Medium (may show 2 peaks for NH_2)	Amines, amides
C-H	2850-3100	Strong	Most organic compounds
C=O	1680-1750	Strong, sharp	Aldehydes, ketones, acids, esters, amides
C=C	1620-1680	Medium	Alkenes
C-O	1000-1300	Strong	Alcohols, esters, ethers

Fingerprint region (below ~1500 cm^{-1}) is unique to each compound; database matching for identification. Greenhouse gases absorb IR because bond vibrations change dipole moment (CO_2, H_2O, CH_4). N_2 and O_2 do not.

Exam Tip: Wavenumbers are in the data booklet. Focus on: broad O-H (alcohol vs very broad acid), sharp C=O, disappearance of O-H after esterification. Compare spectra before/after reactions to confirm functional group changes.

6.11 NMR Spectroscopy

^{13}C NMR

Number of peaks = number of different carbon environments. Peak height does NOT indicate number of carbons. Chemical shifts: alkyl C (5-40 ppm), C-O (50-90), C=C/aromatic (110-160), C=O acid/ester (160-185), C=O aldehyde/ketone (190-220).

^{1}H NMR — Three Key Pieces of Information

1. Chemical shift (δ)

Position relative to TMS (0 ppm). Key values: R-CH_3 (0.7-1.6), R-CH_2-X (3.0-4.2), R-OH (0.5-6.0, variable), R-CHO (9.0-10.0), Ar-H (6.5-8.0), R-COOH (10.0-12.0).

2. Integration

Peak area ∝ number of H atoms in that environment. Given as a ratio.

3. Splitting (n+1 rule)

A proton with n equivalent neighbours on adjacent carbons splits into (n+1) peaks: 0 → singlet, 1 → doublet, 2 → triplet, 3 → quartet.

D_2O Shake

Adding D_2O makes O-H and N-H peaks disappear (proton exchange with deuterium). Identifies exchangeable protons.

Exam Tip: O-H and N-H appear as broad singlets (no splitting due to rapid exchange). In combined-technique questions: MS for M_r, IR for functional groups, ^{13}C NMR for carbon environments, ^{1}H NMR for H environments and connectivity. Degree of unsaturation (DBE) = (2C + 2 + N - H - X)/2; DBE of 4 suggests a benzene ring.

6.12 Chromatography

TLC (Thin Layer Chromatography)

Stationary: silica/alumina plate. Mobile: organic solvent. R_f = distance by substance / distance by solvent front. Quick qualitative identification and purity check.

GC (Gas Chromatography)

Stationary: high-bp liquid in column. Mobile: inert carrier gas. Separates volatile compounds. Retention time identifies components. Peak area gives quantitative composition.

GC-MS

GC separates the mixture; MS identifies each component by molecular mass and fragmentation pattern. Extremely powerful for forensics, environmental monitoring, drug testing, and quality control.

HPLC

Stationary: fine particles in column. Mobile: liquid solvent at high pressure. For non-volatile and thermally unstable compounds (pharmaceuticals, proteins, biological samples).

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